ALI KHAN

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Education

Rutgers Honors College — New Brunswick, NJ

B.S. in Computer Science and Mathematics; GPA: 4.00/4.00 — *Expected May 2028*

Mathematics *Real Analysis I, Linear Algebra, Probability, Combinatorics, Stochastic Calculus, Linear Optimization*

Computer Science *Economics and Computation*, Data Structures and Algorithms*, Artificial Intelligence, Discrete Structures*

**Graduate Coursework*

Manuscripts

A. Khan. "Channels to Targets: A Closed-Form Coordinate System for Silence-Blindness in LLM Observers." Manuscript. Code: github.com/akhan531/Control-On-LLMs *2026*

Research Experience

Research Fellow, Chancellor's Signature Initiative SURF — May – Jul 2026

Rutgers–New Brunswick Office for Research | Faculty mentor: Prof. Laurent Burlion — *New Brunswick, NJ*

- Independently developed a dynamical-systems characterization of multi-agent LLM debate, modeling deliberation as Bayesian belief updating and tracking progress by the divergence between the group's public belief and the full-information posterior
- Established that a Lyapunov argument certifies convergence under honest disclosure and fails under strategic withholding; validated the model against Llama-3.3-70B debates over 40 seeds
- Demonstrated a misaligned incentive in LLM-judge credit assignment: withholding a pivotal clue until late in the debate earns a higher credit share than early disclosure, and the effect strengthens with judge capability

Undergraduate Researcher — Sep 2025 – May 2026

Mathematical Optimization Research Group | Advisors: Prof. Farzad Yousefian, Fateme Maleki — *New Brunswick, NJ*

- Designed IncentOSFL, a one-shot variant of an incentivized federated learning framework, by observing that a convex loss drives every client's local model to the same global optimum, making per-round aggregation redundant
- Reduced per-round communication from full model parameters to data-contribution values alone; validated on MNIST, showing no client drift and convergence of client contributions to a Nash equilibrium independent of initialization
- Built an industrial case study on the UCI Steel Plates Faults dataset demonstrating that the incentivized mechanism outperforms the data-gatekeeper's selfish optimum for every participant
- Supported by U.S. Department of Energy grants #DE-SC0023303 and #DE-SC0025570

Research Fellow, Aresty Summer Science Program — May – Jul 2025

Swarm Intelligence Lab | PI: Prof. Daniel Burbano Lombana | Mentor: Deze Liu — *New Brunswick, NJ*

- Analyzed zebrafish obstacle-avoidance behavior under five illumination conditions using visual-input metrics, extracting trajectories with AnimalTA and filtering position data in MATLAB
- Quantified illumination effects on linear speed, opacity, thigmotaxis, and spatial entropy via factorial analysis across single-fish and dyad conditions
- Trained a random forest classifier with recursive feature selection to distinguish obstacle-avoidance behaviors at 90% accuracy

Selected Projects

Genomic Variant Triage with Evo2 | *Python, Next.js, TypeScript, Modal, Anthropic API* — Feb – Mar 2026

- Built a variant-interpretation tool scoring single nucleotide variants by delta log-likelihood from the Evo2 7B genome foundation model over 8,192 bp UCSC reference windows, served on Modal H100 GPUs
- Designed a splicing-disruption module binning canonical splice-site risk by exon–intron boundary distance from the Ensembl REST API, capturing pathogenicity the raw likelihood score misses
- Developed a VUS triage pipeline ranking panels by a weighted composite of pathogenicity, splice severity, cancer-gene tier, and confidence, with a 29-gene annotation database, live ClinVar context, sentence-transformer RAG, and LLM-generated tumor board notes; covered by 35 unit tests
- Code: github.com/akhan531/Biotech-AI-Tool

Equilibrium in a Universal Basic Income Economy | *Math 490, Prof. Kim Weston* — May 2026

- Modeled a continuous-time market with heterogeneous CARA agents receiving stochastic income transfers, deriving the state-price deflator from martingale conditions on discounted assets

- Solved the agents' dynamic optimization via an HJB equation under an exponential ansatz, obtaining closed-form optimal portfolio and initial consumption
- Imposed stock and initial-consumption market clearing to reduce equilibrium characterization to a PDE system for the agents' value functions

Regime-Robust Signal Weighting | *CS 494 Independent Study, Prof. Amélie Marian* Spring 2026

- Adapted the Disparity Compensation Algorithm from explainable fair ranking to portfolio construction by reframing demographic disparity as a gap in risk-adjusted performance across volatility regimes
- Converted the mechanism from additive selection bonuses to multiplicative signal weights; implemented an end-to-end backtest over 2010–2024 DJIA data with rolling-median regime detection and dollar-neutral portfolio construction
- Code: github.com/akhan531/CS494-Project

Finite State Automata Simulator | *JavaScript, TypeScript, Vis.js* Jul 2025

- Built an interactive deterministic automata simulator with real-time state visualization; deployed to the Rutgers CS Department as an instructional tool supporting 200+ students

Talks and Presentations

"Belief Dynamics and Strategic Timing in Multi-Agent LLM Debate." A. Khan and L. Burlion. Poster and lightning talk, *2026 CSI SURF Research Symposium*, Rutgers–New Brunswick. *Jul 2026*

"IncentOSFL: A Communication-Efficient Variant for Incentivized Federated Learning." A. Khan, F. Maleki, F. Yousefian. Poster, *2026 ISE Research Day*, Rutgers Department of Industrial and Systems Engineering. *May 2026*

"Convex Regularized Portfolio Optimization." Directed Reading Program symposium, Rutgers Department of Mathematics. Mentor: Anupam Nayak. *Dec 2025*

"Vision-based Collective Group Formations in Fish Schools." A. Khan, D. Liu, D. Burbano Lombana. Poster, *2025 Aresty Summer Research Symposium*, Rutgers–New Brunswick. *Jul 2025*

Awards, Honors, and Fellowships

Chancellor's Signature Initiative Summer Undergraduate Research Fellowship, Rutgers–New Brunswick Office for Research. Competitive university-wide research fellowship, $6,000. *2026*

Second Place, Undergraduate Research Poster Presentation, 2026 ISE Research Day, Rutgers Department of Industrial and Systems Engineering. Awarded for work on incentivized federated learning; $200 prize. *2026*

Aresty Summer Science Program, Aresty Research Center for Undergraduates, Rutgers University. Competitive summer research fellowship, $3,000. *2025*

Dean's Scholarship, Rutgers Honors College. Merit scholarship, $5,000 per semester. *2024 – present*

Dean's List, Rutgers School of Arts and Sciences. All semesters. *2024 – present*

Teaching Experience

Part-Time Lecturer Sep 2025 – Present
Department of Computer Science, Rutgers University *New Brunswick, NJ*

- Lead a weekly recitation section of approximately 50 students and hold weekly office hours

CS 205, Introduction to Discrete Structures I | Prof. Akash Sengupta *Fall 2026, Spring 2026*
CS 205, Introduction to Discrete Structures I | Prof. Jingjin Yu *Fall 2025*

Grader Jun 2025 – Present
Department of Mathematics, Rutgers University *New Brunswick, NJ*

Math 300, Introduction to Mathematical Reasoning | Prof. Juan Pablo Mejía-Ramos *Fall 2026*
Math 300, Introduction to Mathematical Reasoning | Prof. Christopher Alexander *Fall 2026*
Math 478, Introduction to Stochastic Processes | Prof. Heeyoung Kwon *Spring 2026*
Math 477, Mathematical Theory of Probability | Prof. Joshua Salamon *Fall 2025*
Math 477, Mathematical Theory of Probability | Prof. Triet Pham *Summer 2025*

Technical Skills

Languages: Python, MATLAB, C++, Java, JavaScript, SQL
Scientific Computing: NumPy, Pandas, Scikit-Learn, CVXPY, Matplotlib | **LLM Infrastructure**: vLLM, Modal
Tools: Git, LaTeX, Jupyter, Mathematica